INTERNATIONAL MONETARY FUND

SAMOA

STAFF REPORT FOR THE 2019 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITY ANALYSIS

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Prepared by the staffs of the International Monetary Fund (IMF) and the International Development Association (IDA)[1]

Risk of external debt distress	*High*
Overall risk of debt distress	*High*
Granularity in the risk rating	*Sustainable*
Application of judgment	*Yes: high probability of a large and persistent breach of the threshold from 2034*

Samoa remains at high-risk of external debt distress under the revised Debt Sustainability Framework (DSF) introduced in July 2018. Consistent with previous analyses, the average long-term costs of natural disasters and climate change are incorporated into the baseline scenario to consider their impact on economic growth, the fiscal position and current account balance. The result is a breach of the threshold under the baseline scenario for the present value of the external public-and publicly guaranteed (PPG) debt-to-GDP ratio from 2036. Stress tests confirm the vulnerability of the debt position to plausible shocks. A tailored natural disaster shock, similar in scale to the median impact of natural disasters in Samoa's history, causes a large and significant deterioration in debt sustainability in the aftermath of the event. A contingent liability shock would also worsen debt sustainability. Given Samoa's vulnerability to natural disasters, strategies to strengthen Samoa's resilience to economic shocks should continue to be implemented. A 'mechanical' application of the new DSF would imply that the risk of debt distress in Samoa is moderate.[2] However, staff judgement has been applied to arrive at a high-risk assessment as there is a high probability the threshold will be largely and persistently breached in the long-run due to Samoa's exposure to climate change despite government efforts to mitigate its impacts. While domestic debt remains small, domestic guaranteed debt accounts for 7 percent of GDP in 2017/18.

[1] This DSA has been prepared jointly by the IMF and World Bank, following the revised LIC-DSF Framework and Guidance Note (2017) in effect as of July 1, 2018. Samoa's debt carrying capacity is classified as strong, based on its Composite Indicator (CI) value of 3.31, based on the October 2018 WEO and updates to the CPIA index through 2017.Thresholds for debt burden indicators are those established in the revised framework.

[2] This is due to a shortening of the period for analysis to a 10-year forecast horizon (from 20 years) under the revised LIC-DSF (Staff Guidance Note).

PUBLIC DEBT COVERAGE

1. Central government and central government-guaranteed debts are covered in this report. Public and Publicly Guaranteed debt accounted for 57.2 percent of GDP in FY2017/18, allocated between central government debt (50.3 percent of GDP) and central government-guaranteed debt (6.9 percent of GDP). The government-guaranteed debt includes SOE debt, which is explicitly guaranteed by central government. The Samoa National Provident Fund, a social security fund, does not owe debt. There is no sub-national government structure in Samoa, nor extra budgetary funds. The Central Bank of Samoa is not allowed to contract debt on behalf of the government. Non-guaranteed SOE debt estimates are not available and cannot be included in PPG debt. The definition of external and domestic debt is based on residency[3]. The authorities are currently working towards strengthening the monitoring and disclosing of fiscal risks from the SOEs with the assistance of PFTAC. The creation of a debt management unit would also improve debt data coverage.

	Subsectors of the public sector	Sub-sectors covered
1	Central government	X
2	State and local government	
3	Other elements in the general government	
4	o/w: Social security fund	X
5	o/w: Extra budgetary funds (EBFs)	
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	
8	Non-guaranteed SOE debt	

1	The country's coverage of public debt	The general government, central bank, government-guaranteed debt			
		Default		Used for the analysis	Reasons for deviations from the default settings
2	Other elements of the general government not captured in 1.	0	percent of GDP	0.0	
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2	percent of GDP	2.0	
4	PPP	35	percent of PPP stock	0.0	
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5	percent of GDP	5.0	
	Total (2+3+4+5) (in percent of GDP)			7.0	

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

BACKGROUND ON DEBT

2. Public debt increased to 50.3 percent of GDP in 2017/18, from 49.1 percent in FY2016/17. The depreciation of the Tala against the main loan currencies, the continuous disbursements of external loans and limited increase in GDP contributed to the increase of the debt-to-GDP ratio.

3. Public external debt was US$421 million at the end of FY2017/18, increasing by 3.6 percent with respect to FY2016/17. The outstanding public external debt at the end of FY2017/18 was SAT1.09 billion, 51 percent from multilateral creditors – including IDA and ADB – and 49 percent from bilateral creditors – including China and Japan. The increase in public external debt and in debt service was driven by the loss of access to exclusive grant funding from IDA and ADB in 2015 and 2016, and on further disbursements of previously contracted loans from bilateral creditors for large infrastructure projects.

[3] Since all the domestic debt is in tala and all the external debt is in foreign currency or SDRs, the residency-based classification is equivalent to the currency-based classification in the case of Samoa.

Samoa: Evolution of Public Debt, FY 2015/16 – FY 2017/18

	FY2015/16	FY2016/17	FY2017/18	FY2015/16	FY2016/17	FY2017/18
	In millions of SAT	In millions of SAT	In millions of SAT	In percent of external debt	In percent of external debt	In percent of external debt
Total public debt	1080.8	1047.3	1113.8	--	--	--
External	1041.8	1018.0	1094.3	100.0%	100.0%	100.0%
Multilateral	546.9	534.5	563.0	52.5%	52.5%	51.5%
IDA	247.3	247.2	280.8	23.7%	24.3%	25.7%
ADB	270.2	256.1	250.6	25.9%	25.2%	22.9%
OPEC	20.7	23.2	24.0	2.0%	2.3%	2.2%
Other	8.7	8.0	7.6	0.8%	0.8%	0.7%
Bilateral	495.0	483.5	531.3	47.5%	47.5%	48.5%
EXIM Bank (China)	413.7	410.0	440.1	39.7%	40.3%	40.2%
Japan	81.3	73.5	91.2	7.8%	7.2%	8.3%
Domestic	38.9	29.4	19.5	--	--	--
Memorandum items:						
Nominal GDP	2055.3	2133.2	2213.5			
Market rate (tala/U.S. dollar, end period)	2.6	2.5	2.6			

Sources: Samoan authorities; and IMF staff estimates.

4. Domestic debt remains below 1 percent of GDP. All domestic public debt is held by domestic creditors and issued in local currency. It is mainly composed by government loans provided by the central government to public entities.

Samoa: Stock of Public Debt at the End of FY 2017/18

	In millions of SAT	In millions of US dollars	As a share of total debt	In percent of GDP
Total public debt	1113.8	433.6	100.0%	50.3%
External	1094.3	426.0	98.2%	49.4%
Multilateral	563.0	219.2	50.5%	25.4%
IDA	280.8	109.3	25.2%	12.7%
ADB	250.6	97.5	22.5%	11.3%
OPEC	24.0	9.4	2.2%	1.1%
Other	7.6	3.0	0.7%	0.3%
Bilateral	531.3	206.8	47.7%	24.0%
EXIM Bank (China)	440.1	171.3	39.5%	19.9%
Japan	91.2	35.5	8.2%	4.1%
Domestic	19.5	7.6	1.8%	0.9%
Memorandum items:				
Nominal GDP	2213.5			
Market rate (tala/U.S. dollar, end period)	2.6			

Sources: Samoan authorities; and IMF staff estimates.

5. Guarantees accounted for 6.9 percent of GDP in FY2017/18, with public and publicly guaranteed (PPG) debt totaling 57.2 percent of GDP. Government-guaranteed debt is held by domestic creditors and accounted for 6.9 percent of GDP in 2017/18, from 7.9 percent in 2016/17.

METHODOLOGY AND ASSUMPTIONS

6. The underlying assumptions are consistent with the macroeconomic framework, based on updated data provided by the authorities and estimates by staff. The baseline scenario incorporates the effects of natural disasters and climate change over the longer term. The years 2019–24 are assumed to be disaster free to simplify the policy discussion of the near-term outlook. From 2025 onwards, the baseline incorporates the average long-term effects of natural disasters and climate change by lowering GDP growth by 1.3 percentage points (pp) annually, raising the current account deficit by 1.5 pp and increasing the fiscal deficit by 1.5 pp vis-à-vis disaster-free projections to reflect the country's historical experience. These are consistent with the findings of staff's analysis on the impact of natural disasters.[4] The discount rate used to calculate the net present value of external debt is 5 percent. The main assumptions are:

- **Real GDP growth** is projected to rebound to 3.4 percent in FY2018/19 driven by commerce, infrastructure spending, and the development of the transport and communication sector. Growth in FY2019/20 is expected to spike to 4.4 percent driven by tourism related to Samoa hosting the Pacific Games (PG) in July 2019, before normalizing at about and 2.2 percent in the medium-term. To account for the average impact of natural disasters, the growth rate is lowered by 1.3 percentage points after 2024.

- After a temporary increase driven by the impact of cyclone Gita on local food prices, a one-time increase in education fees, and higher price on imported fuel, **inflation** is expected to stabilize at around 2.8 percent over the medium term.

- **The current account deficit** is projected to remain below 1 percent of GDP between 2019-2024. The current account recorded a temporary surplus of 2.3 percent of GDP in 2017/18 driven by higher than average remittances, on the aftermath of Cyclone Gita, and tourism receipts. Remittances are expected to normalize from 2018/19, but tourism receipts are projected to remain strong in anticipation of the PG. To account for the average impact of natural disasters, the deficit is widened by 1.5 percentage points after 2024.

Samoa: Baseline Macroeconomic Assumptions
(In percent of GDP, unless otherwise noted)

	2018 DSA		2017 DSA	
	2019-2024	2025-2039	2018-2023	2024-2038
Real GDP growth (in percent change)	2.8	0.9	2.8	0.9
Inflation (change in percent)	2.8	2.8	2.5	3.0
Current account deficit	0.8	5.1	3.7	6.3
Overall fiscal balance deficit	2.2	5.1	2.3	5.0

- **The overall fiscal balance** is projected to return to deficit from FY2017/18. Given recently-legislated increases in public servant wages for cost of living adjustment, the projected reduction in budget support grants and the need to scale up infrastructure projects, staff

[4] See Samoa Article IV consultation, 2017.

projects a 1.2 percent of GDP deficit in 2018/19, which will widen to 2.7 percent in 2023/24. To account for the average impact of natural disasters, the deficit is widened by 1.5 percentage points after 2024.

- **Continued eligibility for concessional borrowing from MDBs** is assumed for the forecast period to finance the fiscal deficit. The grant element of new loans is 40 percent on average. It is assumed that borrowing from the IDA and the Asian Development Bank is on full credit terms[5].

7. The new realism tools suggest that the projections are reasonable. Both external and public PPG debt projections are in line with last year DSA (Figure 3). At the same time factors contributing to debt dynamics remain broadly in line with historical contribution, with the exception of the impact of current account balance and price that is due to temporary factors. The primary balance is expected to be -2.4 percent of GDP in 2020/2021. This belongs to the middle of the distribution of other countries experience. Forecasted real growth in 2019 and 2020 is higher than paths obtained based on plausible fiscal multiplier effects. This is due to the exceptionally low GDP growth in 2017/18, the basis year to calculate the growth path, and to the projection of additional and temporary expected growth in FY2018/19 and FY2019/20 due to the Pacific Games. Public investment has declined in FY2017/18 due to the completion of large infrastructure projects in the previous fiscal year but is expected to rebound in FY2019/20. Public capital stock is expected to contribute to about one third of the GDP growth, in line with historical data and previous DSA (Figure 4).

COUNTRY CLASSIFICATION

8. The country's debt-carrying capacity remains strong, though the indicator has declined. Samoa's Composite Indicator (CI) index, which has been calculated based on the October 2018 WEO and 2017 CPIA, is 3.31, indicating that the country's debt-carrying capacity is high in the revised LIC-Debt Sustainability Framework. The CI is lower than during the previous version of the DSA, but remains associated with a strong rating. The rating is the highest among the Pacific island countries (PICs), and debt is therefore assessed against a higher threshold compared to other PICs.

9. The relevant indicative thresholds for the countries with a strong CI rating are 55 percent for the PV of debt-to-GDP ratio, 240 percent for the PV of debt-to-exports ratio, 21 percent for the debt service-to-exports ratio, and 23 percent for the debt service-to-revenue ratio. These thresholds are applicable to public and publicly guaranteed (PPG) external debt. The benchmark for the PV of total public debt under strong capacity is 70 percent.

[5] With respect to projected new borrowing from IDA and ADB, DSAs always assume terms that would prevail if the country was at low risk of debt distress, independent of current actual terms (which can change on a year to year basis). This is done to avoid a circular situation where the assumption that future commitments will be on grant terms would yield actual commitments on credit terms.

Components	Coefficients (A)	10-year average values (B)	CI Score components (A*B) = (C)	Contribution of components
CPIA	0.385	3.995	1.54	46%
Real growth rate *(in percent)*	2.719	2.566	0.07	2%
Import coverage of reserves *(in percent)*	4.052	32.791	1.33	40%
Import coverage of reserves^2 *(in percent)*	-3.990	10.753	-0.43	-13%
Remittances *(in percent of GDP)*	2.022	15.494	0.31	9%
World economic growth *(in percent)*	13.520	3.660	0.49	15%
CI Score			3.31	100%
CI rating			Strong	

Debt carrying capacity (CI classification)	PV of PPG external debt in percent of		PPG external debt service in percent of		PV of total public debt
	GDP	Exports	Exports	Revenue	GDP
Weak	30	140	10	14	35
Medium	40	180	15	18	55
Strong	55	240	21	23	70

DETERMINATION OF SCENARIO STRESS TESTS

10. Given the severity and frequency of natural disasters in Samoa, a tailored stress test for a natural disaster shock was conducted. Parameters for this test were based on staff research[6] and assume a one-off shock of 21 percent to the debt-GDP ratio in 2019, corresponding to the impact of the median natural disaster shock in the country history, and a reduction of real GDP growth and exports by 3 percent and 6 percent respectively.

11. The stress test for the combined contingent liabilities accounts for implicit liabilities and a potential financial market shock. Default settings of the contingent liability test were used to reflect the likely consequences of a contingent liability shock on the debt path in Samoa. The stress test for contingent liabilities amounts 7 percent of GDP, comprising 2 percent of GDP of non-guaranteed SOE debt and 5 percent of GDP of financial market shock.

[6] Detail in the IMF Working Paper 18/108, "The Economic Impact of Natural Disaster in Pacific Island Countries" (https://www.imf.org/en/Publications/WP/Issues/2018/05/10/The-Economic-Impact-of-Natural-Disasters-in-Pacific-Island-Countries-Adaptation-and-45826).

DEBT SUSTAINABILITY

A. External Debt Sustainability Analysis

12. Under the baseline scenario, Samoa's external debt path is projected to breach the indicative benchmark from 2036 and onward (Figure 1). The PV of debt-to-GDP ratio is expected to increase gradually from 33.4 in 2019 to 39.1 in 2029 and to reach 60.3 in 2039, on the increase of external borrowing to finance construction of infrastructure. The PV of external debt-to-GDP ratio breaches the indicative threshold of 55 percent in 2036. The ratio of the PV of external-debt-to-exports ratio also breaches the threshold of 240 percent in 2036. As a large share of the external debt remains on concessional terms, debt service remains limited and does not breach the indicative threshold.

13. Stress tests confirm the vulnerability of debt dynamics to natural disasters and contingent liabilities. A natural disaster shock has the largest negative impact on the debt trajectory, causing a breach of the threshold for the PV of debt-to-GDP ratio from 2030 (Figure 3). There is a protracted and significant breach of the PV of debt-to-GDP and PV of debt-to-exports after a severe natural disaster shock but also after the contingent liability shock (from 2035 and onward, Table 3).

B. Public Sector Debt Sustainability Analysis

14. Total public debt follows a similar dynamic as the external debt. Under the baseline scenario, the PV of public debt-to-GDP ratio breach the benchmark in 2034 (Table 2). The natural disaster and contingent liability shocks result in a sharper deterioration in debt sustainability. The threshold is breached from 2027 and 2032, respectively (Figure 2).

RISK RATING AND VULNERABILITIES

15. The debt outlook for Samoa is subject to several risks. PPG debt is expected to grow in the medium-term. The external debt-to-GDP ratio is expected to decrease in the short-term on the positive overall fiscal balance registered in FY2017/18 and the expected higher growth in FY2018/19 and FY19/20 due to the Pacific Games. After FY2020/21, the debt ratio is expected to increase again, driven by the negative primary balance. Samoa's external debt breaches the high-risk threshold when the average annual impact of natural disasters and climate change on the medium-term growth, current account, and fiscal balance projections is included in the DSA baseline.

16. Samoa remains at high-risk of debt distress. Another major natural disaster or a materialization of implicit guarantees shock could substantially worsen the debt dynamics. While a mechanical assessment based on the output of the new DSF template would have resulted in a moderate risk assessment, staff's judgement has been applied to conclude on a high-risk assessment as there is a high probability the threshold will be substantially and persistently breached in the long-run due to Samoa's exposure to climate change despite government efforts to mitigate its impacts. The government should continue with recent efforts to consolidate and boost revenue collection, and build fiscal buffers to be prepared to respond to a future economic shock or natural disaster. Introducing a lower public debt-to-GDP target of

40 percent over the long term, associated with operational guidance on the overall deficit, would allow to reduce the level of risk in Samoa. The authorities should also make efforts to widen the coverage of debt data by producing and disseminating aggregate data on general government and public-sector debt (including non-guaranteed SOE debt).

AUTHORITIES' VIEWS

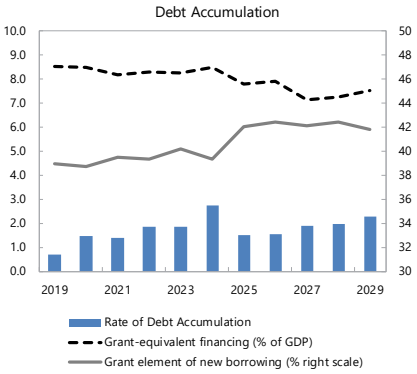
17. The authorities broadly agreed with the DSA assessment and expressed their commitment to reduce the long-term debt target to 40 percent of GDP. The authorities agreed that a lower long-term debt target of 40 percent of GDP (compared to the current target of 50 percent of GDP) is desirable, and would allow them to build fiscal buffers to respond to a large natural disaster, while ensuring adequate funding of development priorities. Consistent with the Medium-Term Debt Strategy, the authorities aim to fund development needs with grants and concessional loans to the extent possible. The authorities are currently working towards strengthening the monitoring and disclosing of fiscal risks from the SOEs with the assistance of PFTAC.

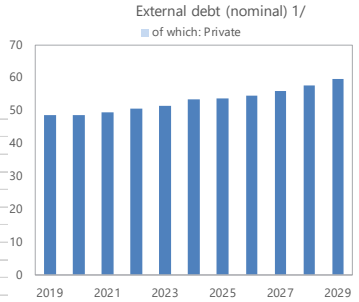
Table 1. Samoa: External Debt Sustainability Framework, Baseline Scenario, 2016-39

(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/	
	2016	2017	2018	2019	2020	2021	2022	2023	2024	2029	2039	Historical	Projections
External debt (nominal) 1/	**50.7**	**47.7**	**49.4**	**48.7**	**48.6**	**49.6**	**50.5**	**51.4**	**53.5**	**59.7**	**87.9**	**46.9**	**53.1**
of which: public and publicly guaranteed (PPG)	**50.7**	**47.7**	**49.4**	**48.7**	**48.6**	**49.6**	**50.5**	**51.4**	**53.5**	**59.7**	**87.9**	**46.9**	**53.1**
Change in external debt	-4.6	-3.0	1.7	-0.7	-0.2	1.0	1.0	0.9	2.1	2.1	1.1		
Identified net debt-creating flows	**5.2**	**-0.5**	**-5.2**	**0.4**	**-2.5**	**-1.1**	**-0.2**	**-0.6**	**-0.1**	**4.2**	**4.1**	**1.5**	**1.5**
Non-interest current account deficit	**4.7**	**1.8**	**-4.0**	**-0.6**	**-0.1**	**-0.3**	**0.2**	**0.2**	**0.5**	**4.2**	**4.1**	**4.3**	**1.9**
Deficit in balance of goods and services	19.2	15.3	14.5	16.8	17.2	17.3	17.5	17.5	17.8	22.0	23.4	**21.4**	**19.4**
Exports	30.7	31.2	32.9	31.3	31.4	31.2	30.9	30.8	30.7	24.3	23.0		
Imports	49.8	46.4	47.4	48.1	48.7	48.5	48.4	48.3	48.5	46.3	46.4		
Net current transfers (negative = inflow)	-16.8	-16.7	-20.2	-20.2	-19.4	-19.7	-19.6	-19.5	-19.4	-20.1	-21.4	**-20.3**	**-19.7**
of which: official	0.0	-0.6	-1.0	-0.4	-0.4	-0.4	-0.4	-0.4	-0.4	-0.4	-0.4		
Other current account flows (negative = net inflow)	2.4	3.2	1.7	2.8	2.1	2.1	2.3	2.2	2.1	2.3	2.0	**3.2**	**2.2**
Net FDI (negative = inflow)	**-0.8**	**1.0**	**-1.7**	**1.5**	**-1.4**	**-0.7**	**0.0**	**-0.3**	**-0.1**	**-0.1**	**-0.1**	**-1.9**	**-0.1**
Endogenous debt dynamics 2/	**1.2**	**-3.3**	**0.6**	**-0.5**	**-1.1**	**0.0**	**-0.4**	**-0.5**	**-0.5**	**0.1**	**0.1**		
Contribution from nominal interest rate	0.0	0.0	1.7	1.1	0.9	1.0	0.6	0.6	0.6	0.6	0.9		
Contribution from real GDP growth	-4.1	-1.3	-0.4	-1.6	-2.0	-1.0	-1.1	-1.1	-1.1	-0.5	-0.8		
Contribution from price and exchange rate changes	5.3	-2.0	-0.7	...	...	...	...	...	...	...	...		
Residual 3/	**-9.8**	**-2.4**	**6.9**	**-1.1**	**2.4**	**2.1**	**1.2**	**1.4**	**2.2**	**-2.1**	**-3.0**	**0.6**	**-0.6**
of which: exceptional financing	2.1	-1.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**...**	**...**	**34.4**	**33.4**	**32.9**	**33.0**	**33.5**	**34.0**	**35.3**	**39.1**	**60.3**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**104.4**	**106.8**	**104.7**	**105.7**	**108.5**	**110.3**	**114.9**	**160.6**	**261.7**		
PPG debt service-to-exports ratio	**...**	**15.2**	**22.3**	**10.6**	**10.1**	**10.5**	**9.7**	**9.4**	**9.2**	**13.3**	**19.1**		
PPG debt service-to-revenue ratio	**...**	**17.5**	**27.7**	**12.3**	**11.6**	**11.9**	**10.9**	**10.6**	**10.3**	**11.7**	**15.9**		
Gross external financing need (Million of U.S. dollars)	31.1	63.2	13.7	38.2	16.4	22.0	33.6	30.4	35.7	95.4	145.2		
Key macroeconomic assumptions													
Real GDP growth (in percent)	7.2	2.7	0.9	3.4	4.4	2.2	2.2	2.2	2.2	0.9	0.9	**1.0**	**2.1**
GDP deflator in US dollar terms (change in percent)	-8.7	4.2	1.5	1.6	1.7	1.6	1.8	1.9	1.8	1.8	2.8	**1.9**	**1.7**
Effective interest rate (percent) 4/	0.0	0.0	3.6	2.3	2.1	2.2	1.3	1.3	1.3	1.1	1.0	**0.4**	**1.5**
Growth of exports of G&S (US dollar terms, in percent)	9.5	8.7	8.2	-0.1	6.6	3.3	2.9	3.8	3.8	2.7	3.8	**4.5**	**1.2**
Growth of imports of G&S (US dollar terms, in percent)	0.1	-0.3	4.5	6.6	7.3	3.6	3.7	3.9	4.5	2.8	3.8	**3.3**	**3.6**
Grant element of new public sector borrowing (in percent)	...	...	...	39.0	38.7	39.5	39.3	40.2	39.4	41.8	35.4	**...**	**40.6**
Government revenues (excluding grants, in percent of GDP)	26.8	27.0	26.4	27.0	27.2	27.4	27.3	27.3	27.3	27.6	27.6	**24.8**	**27.4**
Aid flows (in Million of US dollars) 5/	73.1	60.7	67.7	68.8	70.2	70.8	73.8	78.6	79.8	86.7	64.3		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	8.5	8.5	8.2	8.3	8.3	8.5	7.5	6.4	**...**	**8.0**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	78.2	73.9	72.9	72.1	72.8	68.9	67.4	56.3	**...**	**72.4**
Nominal GDP (Million of US dollars)	786	841	862	905	960	997	1,038	1,080	1,125	1,302	1,738		
Nominal dollar GDP growth	-2.2	7.0	2.4	5.0	6.1	3.8	4.0	4.1	4.1	2.7	3.8	**2.9**	**3.8**
Memorandum items:													
PV of external debt 7/	...	...	34.4	33.4	32.9	33.0	33.5	34.0	35.3	39.1	60.3		
In percent of exports	...	...	104.4	106.8	104.7	105.7	108.5	110.3	114.9	160.6	261.7		
Total external debt service-to-exports ratio	0.0	15.2	22.3	10.6	10.1	10.5	9.7	9.4	9.2	13.3	19.1		
PV of PPG external debt (in Million of US dollars)			296.0	302.3	315.7	329.3	348.0	367.4	397.2	508.6	1047.2		
(PVt-PVt-1)/GDPt-1 (in percent)				0.7	1.5	1.4	1.9	1.9	2.8	2.3	3.6		
Non-interest current account deficit that stabilizes debt ratio	9.4	4.7	-5.7	0.1	0.1	-1.3	-0.8	-0.6	-1.7	2.1	3.0		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + £α (1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, £=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

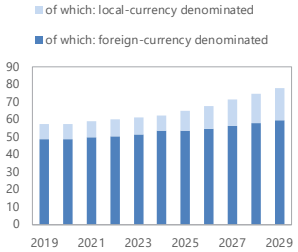
Table 2. Samoa: Public Sector Debt Sustainability Framework, Baseline Scenario, 2016-39

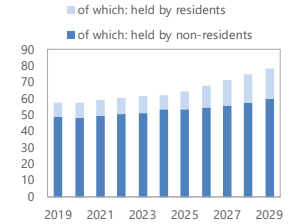
(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2016	2017	2018	2019	2020	2021	2022	2023	2024	2029	2039	Historical	Projections
Public sector debt 1/	**61.3**	**57.0**	**57.2**	**57.5**	**57.6**	**59.2**	**60.3**	**61.4**	**62.5**	**78.2**	**115.9**	**54.1**	**65.0**
of which: external debt	**50.7**	**47.7**	**49.4**	**48.7**	**48.6**	**49.6**	**50.5**	**51.4**	**53.5**	**59.7**	**87.9**	**46.9**	**53.1**
Change in public sector debt	-5.2	-4.4	0.3	0.2	0.1	1.6	1.1	1.1	1.1	3.4	2.9		
Identified debt-creating flows	**-1.5**	**-0.3**	**1.5**	**0.2**	**0.1**	**1.6**	**1.1**	**1.1**	**1.1**	**3.4**	**2.9**	**3.1**	**1.9**
Primary deficit	**0.4**	**1.1**	**-0.1**	**1.2**	**1.8**	**2.3**	**2.5**	**2.7**	**2.7**	**4.5**	**5.5**	**3.6**	**3.2**
Revenue and grants	36.1	34.2	34.3	34.0	33.8	33.6	33.5	33.5	33.3	32.5	31.3	**33.6**	**33.3**
of which: grants	9.3	7.2	7.9	7.0	6.6	6.2	6.2	6.2	6.0	4.9	3.7		
Primary (noninterest) expenditure	36.5	35.3	34.2	35.2	35.7	36.0	36.1	36.3	36.1	37.1	36.9	**37.2**	**36.5**
Automatic debt dynamics	**-1.9**	**-1.4**	**1.6**	**-0.9**	**-1.8**	**-0.7**	**-1.4**	**-1.6**	**-1.6**	**-1.1**	**-2.6**		
Contribution from interest rate/growth differential	-3.0	-1.0	0.7	-1.3	-1.9	-0.9	-1.5	-1.7	-1.7	-1.2	-0.3		
of which: contribution from average real interest rate	1.5	0.6	1.2	0.6	0.5	0.3	-0.2	-0.3	-0.4	-0.5	0.8		
of which: contribution from real GDP growth	-4.5	-1.6	-0.5	-1.9	-2.4	-1.2	-1.3	-1.3	-1.3	-0.7	-1.0		
Contribution from real exchange rate depreciation	1.0	-0.4	0.9	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**-3.7**	**-4.0**	**-1.3**	**0.3**	**0.1**	**0.1**	**0.0**	**0.0**	**0.1**	**0.1**	**-2.4**	**-0.3**	**0.1**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**42.2**	**42.2**	**41.9**	**42.6**	**43.3**	**44.0**	**44.3**	**57.5**	**88.2**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**123.2**	**124.0**	**123.9**	**126.8**	**129.1**	**131.2**	**132.9**	**176.8**	**281.7**		
Debt service-to-revenue and grants ratio 3/	**...**	**18.8**	**28.3**	**17.0**	**17.6**	**15.9**	**13.4**	**11.9**	**11.1**	**13.0**	**16.5**		
Gross financing need 4/	0.4	6.6	9.6	7.0	7.8	7.7	7.0	6.7	6.4	8.8	10.7		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	7.2	2.7	0.9	3.4	4.4	2.2	2.2	2.2	2.2	0.9	0.9	**1.0**	**2.1**
Average nominal interest rate on external debt (in percent)	0.0	0.0	3.6	2.3	2.1	2.2	1.3	1.3	1.3	1.1	1.0	**0.4**	**1.5**
Average real interest rate on domestic debt (in percent)	11.3	7.5	6.5	7.7	5.7	2.5	0.8	-0.3	-0.3	-0.4	-0.5	**1.7**	**1.3**
Real exchange rate depreciation (in percent, + indicates depreciation)	2.0	-0.7	1.9	...	...	...	...	...	...	...	...	**0.2**	**...**
Inflation rate (GDP deflator, in percent)	-1.6	1.1	2.9	2.9	2.7	2.6	2.8	2.9	2.8	2.8	2.8	**1.8**	**2.8**
Growth of real primary spending (deflated by GDP deflator, in percent)	0.4	-0.6	-2.3	6.4	5.7	3.0	2.5	2.8	1.7	0.9	0.9	**3.1**	**2.8**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	5.6	5.4	-0.3	0.9	1.8	0.7	1.4	1.6	1.6	1.1	2.6	**3.6**	**1.3**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Residency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The general government, central bank, government-guaranteed debt. Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

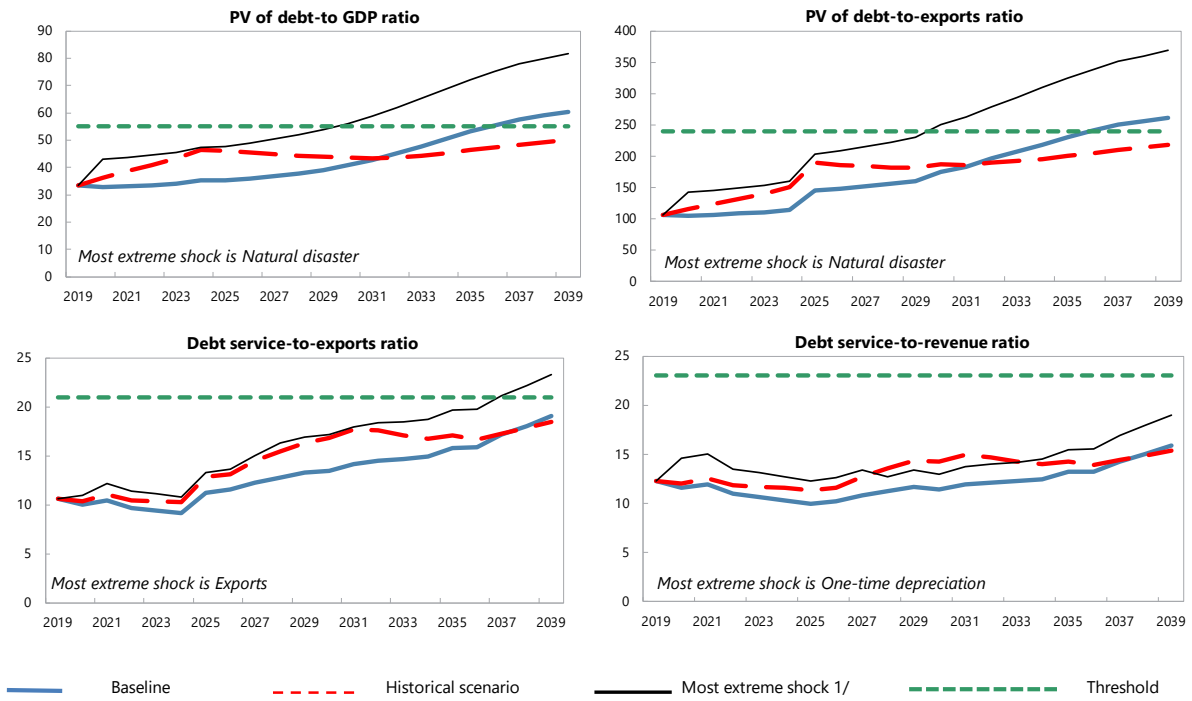
3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 1. Samoa: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2019-39 1/



Customization of Default Settings	Size	Interactions
Standardized Tests	Yes	
Tailored Tests		
Combined CLs	Yes	
Natural Disasters	Yes	Yes
Commodity Prices 2/	n.a.	n.a.
Market Financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	1.1%	1.1%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	24	24
Avg. grace period	6	6

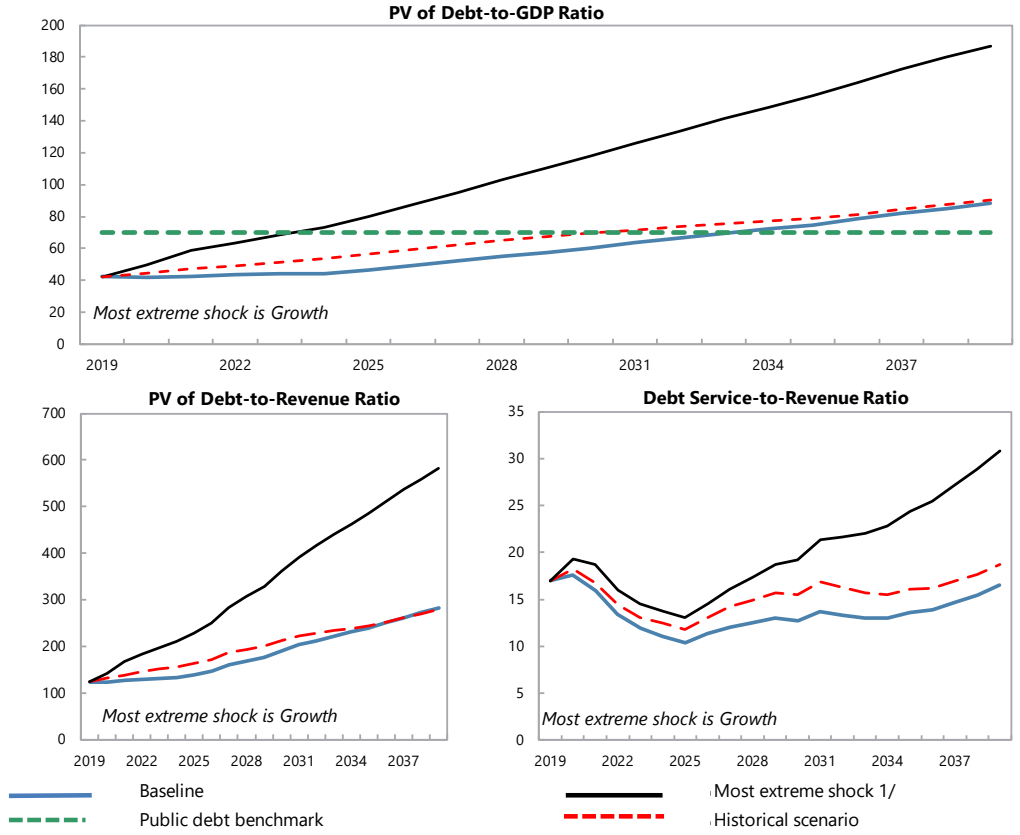
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2029. Stress tests with one-off breaches are also presented (if any), while these one-off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Samoa: Indicators of Public Debt under Alternative Scenarios, 2019-39 1/



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	71%	71%
Domestic medium and long-term	29%	29%
Domestic short-term	4%	0%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	1.1%	1.1%
Avg. maturity (incl. grace period)	24	24
Avg. grace period	6	6
Domestic MLT debt		
Avg. real interest rate on new borrowing	-0.7%	-0.7%
Avg. maturity (incl. grace period)	40	40
Avg. grace period	30	30
Domestic short-term debt		
Avg. real interest rate	-2%	-2%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2029. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Table 3. Samoa: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2019-39

	Projections 1/																				
	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039
PV of debt-to GDP ratio																					
Baseline	33.4	32.9	33.0	33.5	34.0	35.3	35.4	36.0	36.9	37.8	39.1	40.8	42.8	45.2	47.7	50.4	53.1	**55.4**	**57.6**	**58.9**	**60.3**
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2019-2039 2/	33.4	36.2	38.7	40.9	43.3	46.3	46.1	45.4	44.9	44.3	44.1	43.5	43.4	43.7	44.2	45.1	46.3	47.2	48.3	49.2	50.1
B. Bound Tests																					
B1. Real GDP growth	33.4	36.9	40.2	40.8	41.4	43.0	43.1	43.8	44.9	46.0	47.5	49.6	52.1	55.0	**58.0**	**61.3**	**64.6**	**67.4**	**70.1**	**71.7**	**73.3**
B2. Primary balance	33.4	34.7	36.4	36.9	37.4	38.8	38.9	39.5	40.4	41.3	42.5	44.1	46.1	48.4	50.9	53.5	**56.2**	**58.4**	**60.5**	**61.7**	**62.9**
B3. Exports	33.4	34.6	37.5	38.0	38.5	39.8	39.9	40.6	41.4	42.2	43.2	44.7	46.5	48.7	51.0	53.5	**56.1**	**58.1**	**60.1**	**61.3**	**62.4**
B4. Other flows 3/	33.4	35.6	38.4	38.9	39.4	40.7	40.8	41.5	42.3	42.9	43.9	45.3	47.0	49.1	51.3	53.7	**56.2**	**58.2**	**60.1**	**61.1**	**62.1**
B5. Depreciation	33.4	41.4	37.2	37.9	38.4	40.1	40.2	40.9	41.9	43.4	45.1	47.5	50.3	53.6	**57.0**	**60.6**	**64.3**	**67.4**	**70.4**	**72.4**	**74.3**
B6. Combination of B1-B5	33.4	37.8	39.9	40.5	41.0	42.5	42.7	43.4	44.3	45.2	46.5	48.3	50.5	53.1	**55.9**	**58.8**	**61.8**	**64.3**	**66.6**	**68.1**	**69.4**
C. Tailored Tests																					
C1. Combined contingent liabilities	33.4	35.9	36.0	36.6	37.0	38.4	38.5	39.1	40.1	41.1	42.3	44.1	46.2	48.6	51.2	54.0	**56.8**	**59.2**	**61.4**	**62.8**	**64.1**
C2. Natural disaster	33.4	43.1	43.7	44.6	45.5	47.2	47.7	48.9	50.4	51.9	53.7	**56.1**	**58.8**	**61.9**	**65.1**	**68.5**	**72.0**	**75.0**	**77.9**	**79.8**	**81.7**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	55	55	55	55	55	55	55	55	55	55	55	55	55	55	55	55	55	55	55	55	55
PV of debt-to-exports ratio																					
Baseline	106.8	104.7	105.7	108.5	110.3	114.9	145.6	148.0	151.7	155.5	160.6	174.8	183.5	196.2	207.2	218.8	230.8	**240.7**	**250.2**	**256.0**	**261.7**
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2019-2039 2/	106.8	115.2	123.9	132.4	140.6	150.7	189.5	186.5	184.5	182.2	181.1	186.6	186.2	189.6	192.1	195.9	201.3	205.1	209.8	213.7	217.8
B. Bound Tests																					
B1. Real GDP growth	106.8	104.7	105.7	108.5	110.3	114.9	145.6	148.0	151.7	155.5	160.6	174.8	183.5	196.2	207.2	218.8	230.8	**240.7**	**250.2**	**256.0**	**261.7**
B2. Primary balance	106.8	110.4	116.5	119.5	121.4	126.1	159.9	162.6	166.3	169.8	174.6	189.2	197.6	210.2	220.9	232.3	**243.9**	**253.5**	**262.6**	**268.0**	**273.2**
B3. Exports	106.8	119.5	136.7	140.1	142.3	147.7	187.1	190.2	194.3	197.6	202.4	218.4	227.3	**241.1**	**252.7**	**264.9**	**277.6**	**288.0**	**297.8**	**303.3**	**308.8**
B4. Other flows 3/	106.8	113.2	122.9	125.9	127.8	132.5	167.9	170.7	173.9	176.5	180.4	194.2	201.6	213.3	223.0	233.4	**244.1**	**252.8**	**261.0**	**265.5**	**269.9**
B5. Depreciation	106.8	104.7	94.5	97.2	99.0	103.5	131.1	133.3	136.8	141.4	147.3	161.8	171.3	184.7	196.5	208.9	221.7	232.4	**242.7**	**249.4**	**256.0**
B6. Combination of B1-B5	106.8	116.8	112.0	123.7	125.7	130.6	165.5	168.3	171.8	175.4	180.4	195.5	204.4	217.7	229.0	**241.0**	**253.4**	**263.6**	**273.3**	**279.0**	**284.7**
C. Tailored Tests																					
C1. Combined contingent liabilities	106.8	114.2	115.3	118.2	120.1	124.8	158.2	160.8	164.7	168.8	174.1	189.2	198.1	211.3	222.6	234.5	**246.7**	**256.9**	**266.7**	**272.7**	**278.6**
C2. Natural disaster	106.8	142.8	145.4	150.0	153.4	159.7	204.0	208.9	215.3	221.9	229.7	**250.1**	**262.1**	**279.4**	**294.0**	**309.3**	**325.1**	**338.7**	**351.9**	**360.5**	**368.9**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	240	240	240	240	240	240	240	240	240	240	240	240	240	240	240	240	240	240	240	240	240
Debt service-to-exports ratio																					
Baseline	10.6	10.1	10.5	9.7	9.4	9.2	11.3	11.6	12.3	12.7	13.3	13.5	14.1	14.5	14.7	15.0	15.8	15.9	17.2	18.1	19.1
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2019-2039 2/	10.6	10.4	11.0	10.4	10.4	10.3	12.9	13.1	14.5	15.4	16.3	16.8	17.7	17.6	17.1	16.8	17.1	16.7	17.3	17.8	18.4
B. Bound Tests																					
B1. Real GDP growth	10.6	10.1	10.5	9.7	9.4	9.2	11.3	11.6	12.3	12.7	13.3	13.5	14.1	14.5	14.7	15.0	15.8	15.9	17.2	18.1	19.1
B2. Primary balance	10.6	10.1	10.6	9.9	9.6	9.4	11.5	11.8	13.0	14.0	14.5	14.7	15.4	15.8	15.9	16.2	17.1	17.1	18.4	19.3	20.3
B3. Exports	10.6	11.0	12.2	11.4	11.1	10.8	13.3	13.7	15.0	16.4	17.0	17.2	17.9	18.3	18.5	18.8	19.7	19.8	**21.2**	**22.2**	**23.3**
B4. Other flows 3/	10.6	10.1	10.6	10.0	9.7	9.5	11.6	11.9	13.4	14.7	15.2	15.4	16.0	16.4	16.5	16.7	17.5	17.6	18.8	19.6	20.5
B5. Depreciation	10.6	10.1	10.5	9.5	9.2	9.0	11.0	11.4	12.1	11.5	12.1	12.2	12.9	13.3	13.5	13.8	14.7	14.8	16.1	17.1	18.1
B6. Combination of B1-B5	10.6	10.6	11.4	10.5	10.2	10.0	12.3	12.6	14.0	14.5	15.0	15.3	15.9	16.4	16.5	16.8	17.7	17.8	19.1	20.1	**21.1**
C. Tailored Tests																					
C1. Combined contingent liabilities	10.6	10.1	10.6	9.8	9.6	9.3	11.5	11.8	12.5	12.9	13.5	13.7	14.3	14.7	14.9	15.2	16.0	16.1	17.3	18.3	19.2
C2. Natural disaster	10.6	10.8	11.8	10.9	10.7	10.4	12.8	13.2	13.9	14.4	15.1	15.3	16.0	16.5	16.7	17.0	17.9	18.0	19.4	20.4	**21.4**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21
Debt service-to-revenue ratio																					
Baseline	12.3	11.6	11.9	10.9	10.6	10.3	9.9	10.2	10.8	11.2	11.7	11.4	11.9	12.1	12.2	12.5	13.2	13.3	14.3	15.1	15.9
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2019-2039 2/	12.3	12.0	12.6	11.8	11.7	11.6	11.3	11.6	12.7	13.6	14.3	14.2	14.9	14.7	14.3	14.0	14.3	13.9	14.4	14.9	15.4
B. Bound Tests																					
B1. Real GDP growth	12.3	13.0	14.5	13.3	12.9	12.6	12.1	12.4	13.2	13.7	14.3	13.9	14.5	14.7	14.9	15.2	16.1	16.1	17.4	18.3	19.3
B2. Primary balance	12.3	11.6	12.0	11.2	10.8	10.5	10.1	10.4	11.5	12.3	12.8	12.4	13.0	13.1	13.2	13.5	14.2	14.3	15.3	16.1	16.9
B3. Exports	12.3	11.7	12.2	11.3	11.0	10.7	10.3	10.6	11.6	12.6	13.1	12.7	13.3	13.4	13.5	13.7	14.4	14.5	15.5	16.2	17.0
B4. Other flows 3/	12.3	11.6	12.1	11.3	11.0	10.7	10.2	10.5	11.8	12.9	13.4	13.0	13.5	13.6	13.7	13.9	14.6	14.6	15.6	16.4	17.1
B5. Depreciation	12.3	14.6	15.0	13.5	13.1	12.7	12.2	12.6	13.4	12.7	13.4	13.0	13.7	14.0	14.2	14.5	15.4	15.6	16.9	17.9	19.0
B6. Combination of B1-B5	12.3	12.6	13.7	12.6	12.3	11.9	11.4	11.8	13.0	13.5	14.0	13.6	14.3	14.4	14.6	14.8	15.6	15.7	16.9	17.7	18.6
C. Tailored Tests																					
C1. Combined contingent liabilities	12.3	11.6	12.1	11.1	10.8	10.5	10.1	10.4	11.0	11.4	11.9	11.6	12.1	12.3	12.4	12.6	13.4	13.4	14.4	15.2	16.0
C2. Natural disaster	12.3	11.6	12.5	11.6	11.2	10.9	10.5	10.8	11.5	11.9	12.4	12.1	12.6	12.8	13.0	13.2	14.0	14.0	15.1	15.9	16.7
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23	23

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Samoa: Sensitivity Analysis for Key Indicators of Public Debt, 2019-39

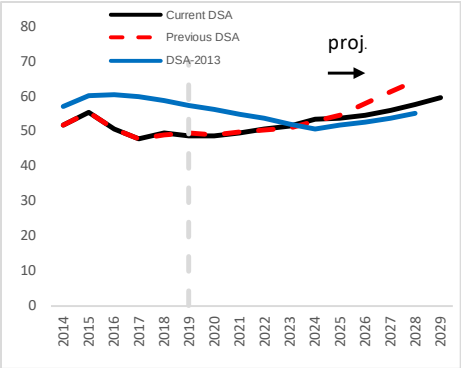
	Projections 1/																				
	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039
PV of Debt-to-GDP Ratio																					
Baseline	42.2	41.9	42.6	43.3	44.0	44.3	46.5	49.2	52.2	54.9	57.5	60.4	63.5	66.5	69.3	**72.1**	**74.8**	**78.3**	**81.8**	**85.1**	**88.2**
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2019-2039 2/	42	45	47	49	52	53	56	59	62	65	67	69	**72**	**73**	**75**	**77**	**79**	**81**	**85**	**88**	**91**
B. Bound Tests																					
B1. Real GDP growth	42	49	59	64	68	**73**	**80**	**87**	**95**	**103**	**110**	**118**	**126**	**134**	**141**	**149**	**156**	**164**	**172**	**180**	**187**
B2. Primary balance	42	45	48	49	50	50	52	55	58	60	63	66	69	**72**	**75**	**77**	**80**	**83**	**87**	**90**	**93**
B3. Exports	42	43	47	47	48	48	51	53	56	59	61	64	67	69	**72**	**75**	**77**	**80**	**84**	**87**	**90**
B4. Other flows 3/	42	45	48	49	49	50	52	55	58	60	62	65	68	**70**	**73**	**75**	**78**	**81**	**84**	**87**	**90**
B5. Depreciation	42	49	48	46	45	43	43	44	44	45	45	46	47	48	49	49	50	52	54	55	57
B6. Combination of B1-B5	42	47	49	50	51	52	55	59	63	66	70	**74**	**77**	**81**	**85**	**88**	**92**	**96**	**100**	**104**	**108**
C. Tailored Tests																					
C1. Combined contingent liabilities	42	47	48	48	49	49	51	54	57	60	62	65	68	**71**	**74**	**77**	**80**	**83**	**87**	**90**	**93**
C2. Natural disaster	42	59	60	61	62	63	66	69	**73**	**76**	**80**	**83**	**87**	**91**	**95**	**98**	**102**	**106**	**110**	**114**	**118**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Public debt benchmark	70	70	70	70	70	70	70	70	70	70	70	70	70	70	70	70	70	70	70	70	70
PV of Debt-to-Revenue Ratio																					
Baseline	124.0	123.9	126.8	129.1	131.2	132.9	138.4	146.4	160.4	169.0	176.8	190.5	202.9	212.3	221.4	230.3	238.8	250.1	261.4	271.7	281.7
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2019-2039 2/	124	131	138	145	151	156	163	171	186	194	200	213	222	228	233	238	243	251	261	270	279
B. Bound Tests																					
B1. Real GDP growth	124	143	168	182	197	211	228	250	284	306	328	362	392	416	440	463	485	511	537	559	582
B2. Primary balance	124	133	144	146	148	149	155	163	177	186	193	207	220	229	238	247	255	266	278	288	297
B3. Exports	124	128	139	141	143	145	151	159	173	181	188	201	213	222	230	238	246	257	267	277	286
B4. Other flows 3/	124	132	143	145	147	149	155	163	177	185	192	205	216	225	233	241	249	259	269	279	288
B5. Depreciation	124	148	144	139	135	130	130	132	138	140	142	147	152	154	157	159	161	167	173	179	184
B6. Combination of B1-B5	124	138	142	147	151	155	163	173	191	202	212	229	245	256	268	279	290	303	317	329	341
C. Tailored Tests																					
C1. Combined contingent liabilities	124	139	142	144	146	147	153	161	175	184	192	206	219	228	237	246	255	266	278	288	298
C2. Natural disaster	124	173	177	181	184	188	194	204	223	233	244	261	277	289	301	312	323	337	351	364	376
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio																					
Baseline	17.0	17.6	15.9	13.4	11.9	11.1	10.3	11.4	12.1	12.5	13.0	12.7	13.7	13.3	13.0	13.0	13.6	13.9	14.7	15.4	16.5
A. Alternative Scenarios																					
A1. Key variables at their historical averages in 2019-2039 2/	17	18	17	14	13	12	12	13	14	15	16	16	17	16	16	16	16	16	17	18	19
B. Bound Tests																					
B1. Real GDP growth	17	19	19	16	15	14	13	15	16	17	19	19	21	22	22	23	24	25	27	29	31
B2. Primary balance	17	18	16	14	12	11	11	12	13	14	14	14	15	14	14	14	15	15	16	17	18
B3. Exports	17	18	16	14	12	11	11	12	13	14	14	14	15	14	14	14	15	15	16	16	17
B4. Other flows 3/	17	18	16	14	12	11	11	12	13	14	14	14	15	15	14	14	15	15	16	17	18
B5. Depreciation	17	18	18	16	14	13	12	14	14	14	15	14	15	14	14	14	14	14	15	16	17
B6. Combination of B1-B5	17	18	17	14	13	12	11	12	13	14	14	14	15	15	14	15	15	16	16	17	18
C. Tailored Tests																					
C1. Combined contingent liabilities	17	18	16	14	12	11	11	12	12	13	13	13	14	14	13	13	14	14	15	16	17
C2. Natural disaster	17	18	17	15	13	12	11	13	13	14	14	14	15	15	14	14	15	15	16	17	18
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the benchmark.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.

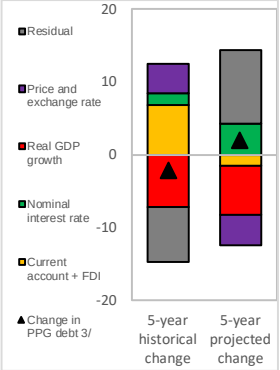
Figure 3. Samoa: Drivers of Debt Dynamics – Baseline Scenario

External Debt

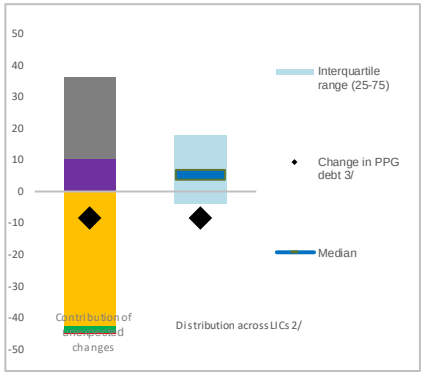
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

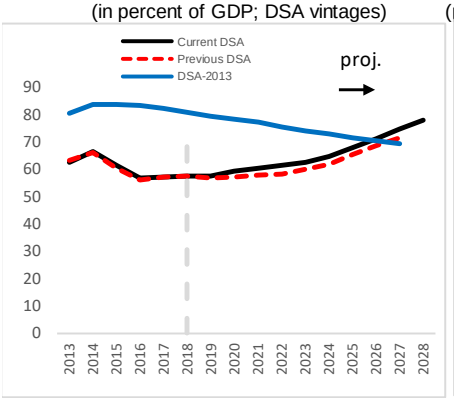


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

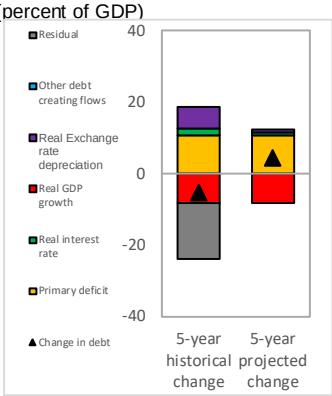


Public debt

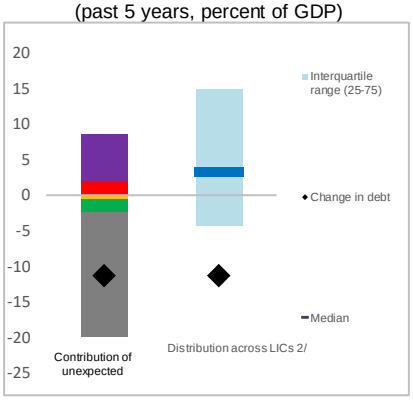
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



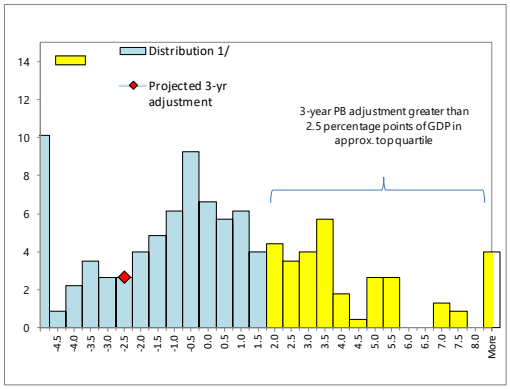
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

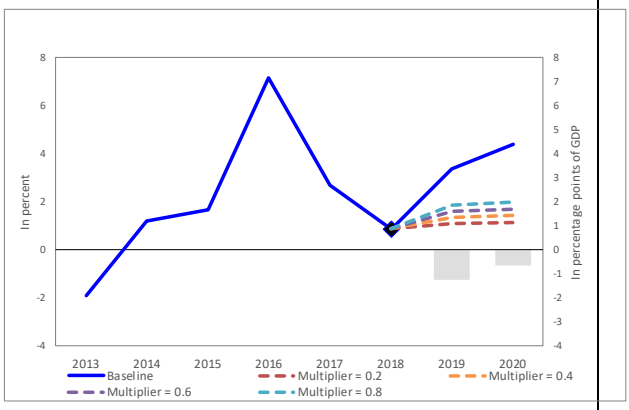
Figure 4. Samoa: Realism Tools

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



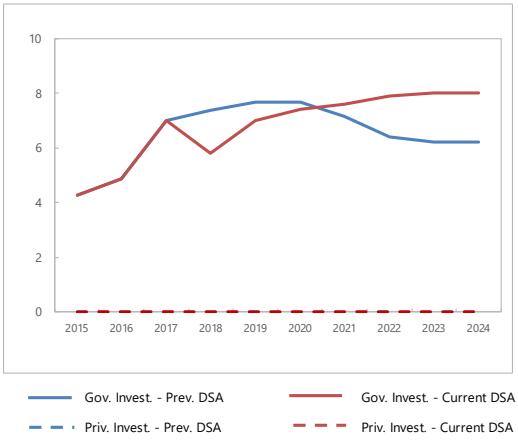
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/

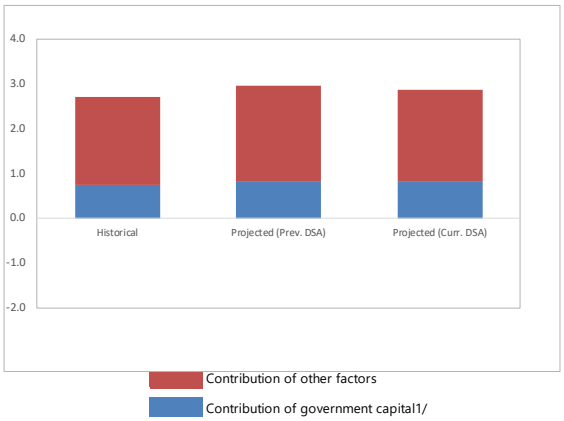


1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(% of GDP)



Contribution to Real GDP growth
(percent, 5-year average)



1/ Capital stock in 2010 corresponds to PFTAC estimates. Later data are obtained by using the net acquisition of non-financial asset growth.